

14.1 A 22nm 104.5TOPS/W μ -NMC- Δ -IMC Heterogeneous STT-MRAM CIM Macro for Noise-Tolerant Bayesian Neural Networks

Publisher: IEEE [Cite This](#)

<< Results

[De-Qi You](#) ; [Win-San Khwa](#) ; [Bo Zhang](#) ; [Fang-Yi Chen](#) ; [Andrew Lee](#) ; [Yu-Cheng Hung](#) 

14

Cites in
Papers

3181

Full
Text Views

Abstract

Authors

Figures

References

Citations

Keywords

Metrics

More Like This

Download PDF

Abstract:

Compute-in-memory (CIM) macros [1]–[5] for convolutional neural networks (CNNs) [6]–[7] and vision transformers (ViTs) [8] enable high-performance computing for energy-ef... [Show More](#)

Metadata

Recommended for You (Beta)

FlexDCIM: A 400 MHz 249.1 TOPS/W 64 Kb Flexible Digital Compute-in-Memory SRAM Macro for CNN...

Algorithm Hardware Co-Design for ADC-Less Compute In-Memory...

Reconfigurable Precision INT4-8/FP8 Digital Compute-in-Memory Macro for AI...

[Learn More](#)

Contents

Compute-in-memory (CIM) macros [1]– [5] for convolutional neural networks (CNNs) [6]– [7] and vision transformers (ViTs) [8] enable high-performance computing for energy-efficient edge-AI devices. Image recognition applications are constrained by inference accuracy degradation or misjudgments due to environmental noise. Bayesian neural networks (BNNs) [9], which represent weights using its mean (μ -weight) and difference from mean tF Δ -weight) outperform CNN and ViT models in terms of noise tolerance, making them promising candidates for edge-AI devices dealing with noisy inputs. Using digital circuits to implement BNN [10]– [12] imposes a tradeoff between inference accuracy and performance, power and area (PPA).

Combining the CIM structure with STT-MRAM is a promising approach for implementing BNNs, thanks to STT-MRAM's stochastic behavior during write operations, its high endurance, and nonvolatility. As shown in [Fig. 14.1.1](#), implementing a BNN using an STT-MRAM CIM imposes several challenges: (1) The exploration of CIM structures - near-memory-compute (NMC) and in-memory-compute (IMC) - is essential for handling MAC computation with μ/Δ -weight and optimizing the tradeoff between PPA and accuracy; (2) The need for accurate and low-power write operations in STT-MRAM devices is critical due to small write-monitor sense margins (WM-SMs) caused by low TMR, especially when using write stochastics tor Δ -weight generation under a given normal, $N(\mu, \sigma)$, distribution; and (3) The degradation of IMC sense margins, caused by near-far effects on BL and a high number of activated WLs, significantly impacts IMC readout accuracy.

This paper proposes the following solutions: (1) A heterogeneous CIM structure using NMC for μ -compute and IMC for Δ -compute ($\mu N - \Delta I$) to achieve high energy efficiency and inference accuracy; (2) A self-compare write-termination (SCWT) scheme to enhance WM-SMs for accurate Δ -weight generation with a minimal PPA impact; and (3) 2D clamping-voltage scaling with sense-margin compensation (2D-CVS-SMC) that improves IMC readout accuracy while reducing power consumption. The test chip, fabricated using foundry-provided 22nm STT-MRAM, achieves 1.045 TOPS/W under an 8bIN – 8b μ – 8b Δ – 27bOUT configuration. In the presence of normalized RGB noise, $\sigma = 0.01$, the inference accuracy degradation of the $\mu N - \Delta I$ STT-MRAM CIM is $2.55 \times$ lower than a simulated CNN.

[Figure 14.1.2](#) illustrates the process for generating BNN weights using the $\mu N - \Delta I$ STT-MRAM CIM macro with the SCWT scheme. The BNN weights, $N(\mu, \sigma)$'s, are generated through offline training. In [Figure 14.1.2](#), $576N(\mu, \sigma)$'s are considered as an example, with 576 μ -weights being directly stored in the μ -array. The user utilizes STT-MRAM write stochastics to generate corresponding 576 Δ – weights. After chip fabrication, the user first establishes a switching-probability (p) table for the STT-MRAM corresponding to the write voltage (V_w) and V_w pulse width (PW). The user then uses this table and the desired σ to generate the corresponding m-bit PW-code via software, which is input into the STT-MRAM stochastic Gaussian random number generator (SS-GRNG). The SS-GRNG consists of a PW generator, an FSM, and an SCWT circuit. The PW generator converts an m-bit PW-code into the V_w PW, which corresponds to a specific p . This PW is provided to the SCWT for central-limit theorem (CLT)-based Δ -weight generation [13]. SCWT tracks near-far effects on BL and PVT variations without the need for an on-chip reference voltage or current. It samples the write cell current (I_{SAMP}) at the onset of the PW (before the STT-MRAM switches) and waits for a change in $I_{DL}(\Delta I_{DL})$. If $\Delta I_{DL} \neq 0$, then the STT-MRAM cell has switched, and SCWT terminates the write process. This operation is repeated N_0 times, with the total switch count X used to calculate Δ – weight = $X - N_0 \times p$. The Δ -weights are then stored in the Δ -array. The user can regenerate a new set of Δ -weights as needed for different applications.

[Figure 14.1.3](#) illustrates the proposed $\mu N - \Delta I$ scheme. Preliminary analysis reveals that the μ -compute error (u-error) has a significant impact on inference accuracy, whereas Δ -compute error (Δ -error) has a negligible impact. Thus, we mapped μ -compute to NMC (more accurate) and Δ -compute to IMC (more energy-efficient). The $\mu N - \Delta I$ architecture uses a custom input register (IN – DFF) to reduce hardware area and power consumption; it also uses a dedicated weight mapping scheme to reduce latencies associated with heterogeneous computation between NMC and IMC. During computation, μ -NMC operates using word-wise inputs and bit-wise weights with only one WL activated at a time. This enables the completion of an 8bIN – 1b μ MAC in 1 cycle; thus, an 8bIN – 8b μ MAC requires 8 cycles. Conversely, Δ -IMC uses a bit-wise inputs, which activates multiple WLs simultaneously, and word-wise weights, enabling the completion of a 1bIN – 8b Δ MAC in 1 cycle; an 8bIN – 8b Δ MAC requires 8 cycles. Note that the μ -NMC and Δ -IMC share a $72 \times 8b$ custom IN-OFF. For the 72 – DL μ -NMC, each OL is mapped to an 8b input, fully utilizing the custom IN-OFF. On the other hand, Δ -IMC uses a 6b selection signal (IN-SEL [5:0]) to select 9 consecutive bit-wise inputs ($IN_{0-8}[k]$), sending it to 9 WLs for up to 9 BL accumulations. For instance, when performing a NN-layer computation, with 64 channels and 576 accumulations, the 8 μ -NMCs ($72DL/bank \times 8banks = 576b$ wide-10) concentrate throughput entirely on a single channel MAC, producing a complete channel output every 8 cycles. This configuration enables the completion of 64-channel outputs in 512 cycles. Meanwhile, the 8 Δ -IMCs (9 BL accumulations, 8 MAC-channel/bank $\times$ 8banks = 64 MAC-channels) distribute throughput evenly across the 64-channel MACs, allowing for the simultaneous completion of 64-channel outputs in 512 cycles. Finally, the u – Δ combiner integrates and outputs 64 sets of μ -MAC and Δ -MAC results.

[Figure 14.1.4](#) illustrates the proposed 2D-CVS-SMC scheme in Δ -IMC: consisting of 2D clamping-voltage scaling (2D-CVS) and 2D sense-margin compensation (2D-SMC) to enhance MAC sense margins (SM) and reduce AOC power consumption during MAC computation. Unlike the scheme in [14], which adjusts the clamping-voltage (V_{CLP}) based only on the WL address (XA), 2D-CVS adjusts V_{CLP} in two dimensions, based on XA and the number of activated WLs (N_{WL}). The 2D adjustment reduces the average BL current and improves MAC energy efficiency (EF_{MAC}), while not degrading the worst-case SM. Following chip fabrication, the 2D-CVS establishes a 2D – V_{CLP} scaling table for each Δ -array; where, a 2b XA [7:6] and a 4b N_{WL} (representing the number of logic-1s in $IN_{0-8}[k]$) correspond to a 3b bias index. This index is stored in the STT-MRAM and is accessed on wake up, where it selects 1 of 8 V_{CLP} levels ($V_{CLP}[7:0]$) to be V_{2D-CLP} . For example, for the case (case-1) where XA is far and N_{WL} is large, the bias index is larger, corresponding to a larger V_{2D-CLP} ; and for case-2, where XA is near and N_{WL} is small, the bias index is smaller, resulting in a smaller V_{2D-CLP} . The V_{2D-CLP} selection operation is performed during WL

development so that it does not contribute to overall latency. The 2D-SMC operation is divided into four phases (PHO - 3): in PHO (standby, overlapping with 2D-CVS): C1 is charged to V_{DD} , while XO and X1 are discharged to V_{SS} . In PH1 (current sampling): SW1 is turned on to sample the MAC current (I_{MAC}) from the Δ -array. C1 records the voltage that mirrors I_{MAC} from P1 to P2. In PH2 (distortion catch): SW1 is turned off, while SW2 and SW3 are turned on. V_{XO} is determined by I_{MAC} mirrored from P2 to P3 and the duration for which SW2 is on. A larger MAC value (MACV) results in a larger I_{MAC} , and thus a larger V_{XO} . The duration of SW2 can be adjusted according to test results to mitigate process variation. V_{X1} is determined by the difference between the 2D reference current ($I_{2D-REFO}$) generated from the Δ -array (with the same XA and N_{WL}) and I_{N4} . $I_{2D-REFO}$ tracks near-far effects on BL, so if $I_{2D-REFO}$ is smaller at a far XA, V_{X1} is smaller, leading to a larger gate-to-source voltage for N5 (V_{XO-X1}), and vice versa. In PH3 (margin compensation and offset cancellation): SW2 is turned off. The compensation current ($I_{2D-BOOST}$) determined by V_{XO-X1} is mirrored from P4 to P5. A current subtractor then cancels the offset currents of $I_{2D-REFO}$, resulting in $I_{2D-OUT} = (I_{MAC} + I_{2D-BOOST} - I_{2D-REFO})$. With a smaller sense current, I_{2D-OUT} , the AOC power consumption is consequently reduced.

[Figure 14.1.5](#) illustrates the simulated performance of the proposed schemes. SCWT enhances WM-SM by 3.25 to $3.74\times$ and reduced write energy by 2.46 to $3.11\times$, compared to both the baseline and conventional write termination [15] schemes. $\mu N - \Delta I$ decreases macro-MAC power by 43% with an inference accuracy loss of only 0.05% , compared to $\mu - \Delta$ all NMC. 2D-CVS-SMC improves IMC EF_{MAC} by $2.48\times$ and MAC SM by $1.90\times$, compared to conventional static-BL clamping. The overall macro-level EF_{MAC} increases by $1.72\times$ with only a 0.03% inference accuracy loss, compared to $\mu - \Delta$ all NMC.

[Figure 14.1.6](#) presents measurement results from the 22nm STT-MRAM CIM macro and a comparison table to prior silicon-verified works. When the ResNet-20 model is applied to the CIFAR-100 dataset, the measured inference accuracy degradation is 1.19% with normalized RGB noise $N(0,0.01)$, which is $2.55\times$ lower than the simulated CNN. The Shmoo plot for the STT-MRAM CIM macro confirms an 18ns computing latency (T_{CD-MAC}) from a 0.8V supply using an 8bIN - 8b μ - 8b Δ - 27bOUT configuration. [Figure 14.1.7](#) presents the die photo and a summary table of key metrics.



Figure 14.1.1:

Advantages of BNN software and challenges of hardware implementation.



Figure 14.1.2:

Bnn-weight-generation process flow within the $\mu N - \Delta I$ stt -mram cim macro using scwt.



Figure 14.1.3:

Bnn-mac computation flow within the $\mu N - \Delta I$ stt-mram cim macro.



Figure 14.1.4:

IMC readout with 2D-CVS-SMC.



Figure 14.1.5:

Simulated performance of the proposed schemes.



Figure 14.1.6:
Measurement results and comparison table to prior works.



Figure 14.1.7:
Die photo and key-metric summary table.

ACKNOWLEDGEMENT

The authors would like to thank NTHU-TSMC JDP, NSTC, and TSRI for financial and manufacturing support.